

Blackbody Radiation

Introduction:

The spectrum of an incandescent light bulb is scanned by hand using a prism spectrophotometer that measures relative light intensity as a function of angle. A Broad Spectrum Light Sensor is used with a prism so the entire spectrum from approximately 400 nm to 2500 nm can be scanned without the overlapping orders caused by a grating. The wavelengths corresponding to the angles are calculated using the equations for a prism spectrophotometer. The relative light intensity can then be plotted as a function of wavelength as the spectrum is scanned, resulting in the characteristic blackbody curve. The intensity of the light bulb is increased, increasing the temperature, and the scan is repeated to show how the curves nest with a shift in the peak wavelength.

The temperature of the filament of the bulb can be estimated indirectly by determining the resistance of the bulb from the measured voltage and current. From the temperature, the theoretical peak wavelength can be calculated and compared to the measured peak wavelength.

Theory:

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The intensity per wavelength, $I_\lambda(\lambda, T)$, as a function of wavelength of radiation emitted by an ideal body (a blackbody since an ideal emitter must also be an ideal absorber) is given by Planck's Radiation Law:

$$E = hf \quad (1)$$

where c is the speed of light in a vacuum, h is Planck's constant, k is Boltzmann's constant, T is the absolute temperature of the body, and λ is the wavelength of the radiation. Any real object must emit less at all wavelengths.

The wavelength with the greatest intensity is given by

$$\lambda_{\max} = (\text{constant})/T = (0.002898 \text{ m}\cdot\text{K})/T \quad (2)$$

The temperature of the blackbody light filament can be calculated using the resistance of the filament while it is lit. We find the resistance (R) by measuring the voltage (V) and the current (I) and using $R = V/I$. The resistance of the tungsten filament is a nonlinear function of the temperature.

The wavelength is determined by measuring the angle at which the light is dispersed by a prism.

Setup:

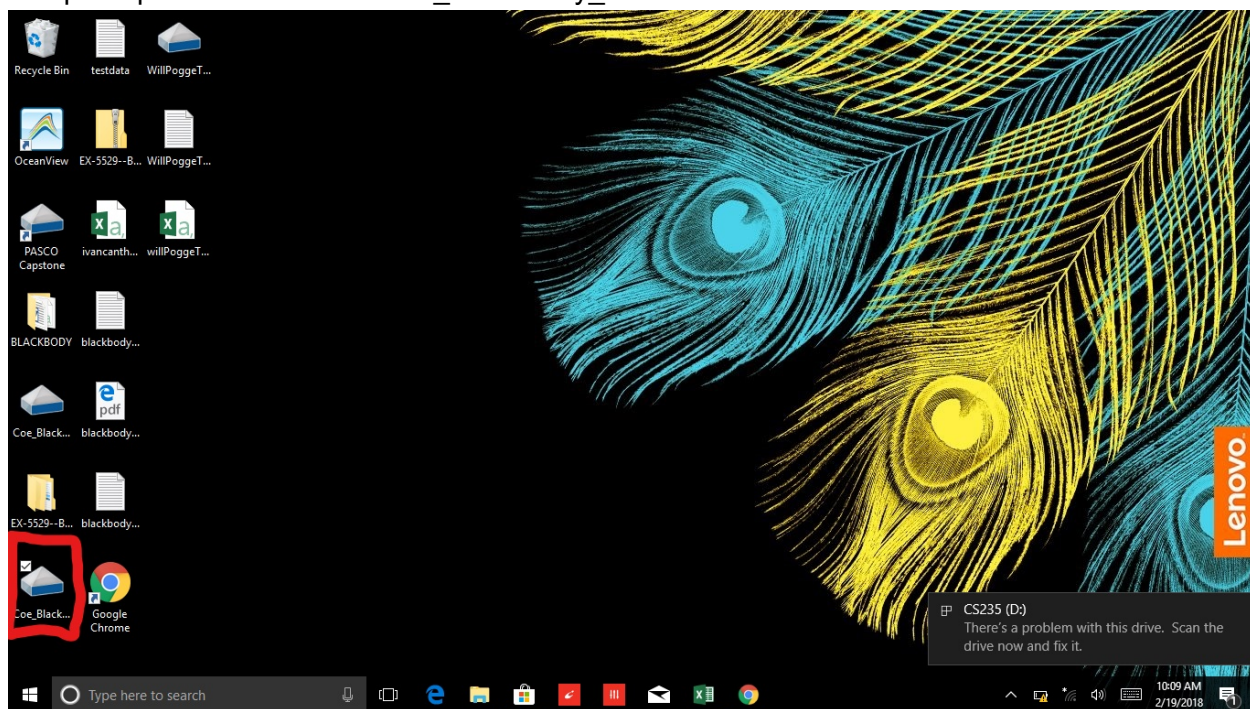
1. Place the Collimating slits on the track 10 cm from the Collimating lense. The slits should be on the #4.

2. Mount the light source to the track as close to the slits as possible

3. Connect the Light Source to the Power Amplifier. (Order doesn't matter)
4. Connect the power amplifier to slot A on the ScienceWorkshop.
5. Connect the cords from the angle sensor to the 1 (yellow) & 2 (black) slots on the ScienceWorkshop.
6. Connect the cord from the Light Sensor to slot b on the ScienceWorkshop.
7. Connect the cord from the back of the Power Amplifier to slot C on the ScienceWorkshop.
8. Plug the USB from the ScienceWorkshop into the computer.

Experiment

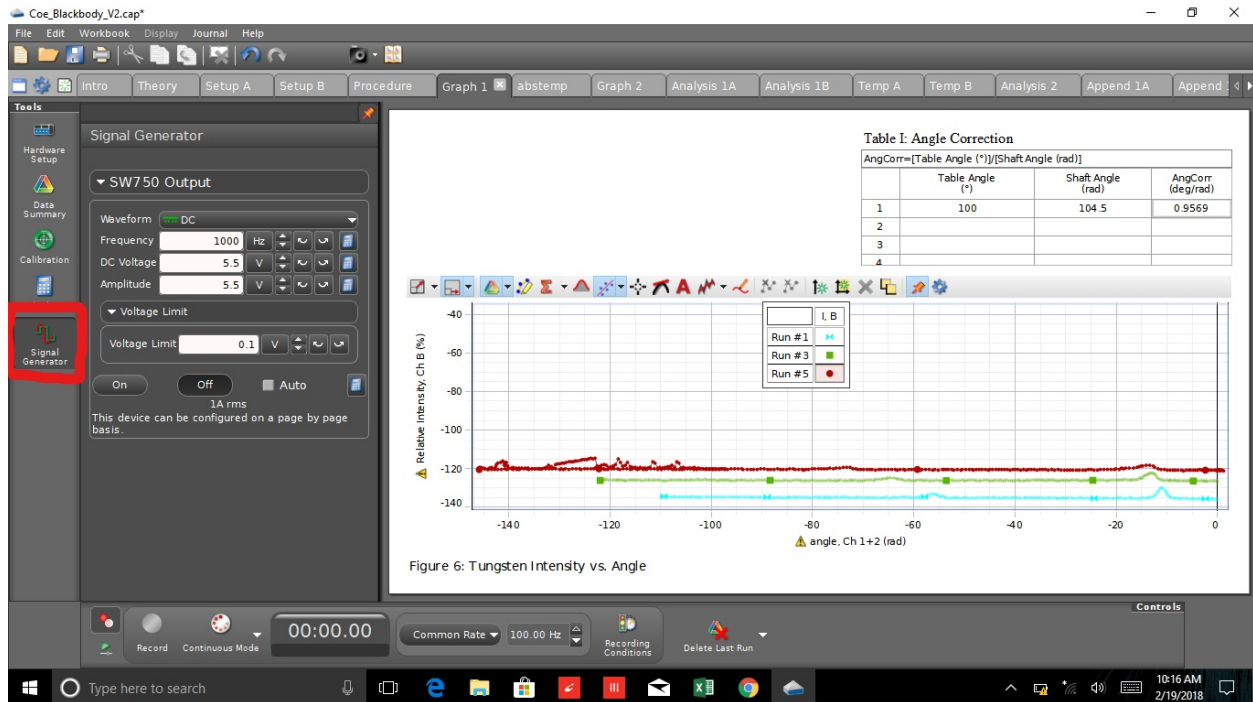
1. Open up the CAPSTONE Coe_BlackBody_V2 Icon on the home screen.



2. On the tab bar select graph1.

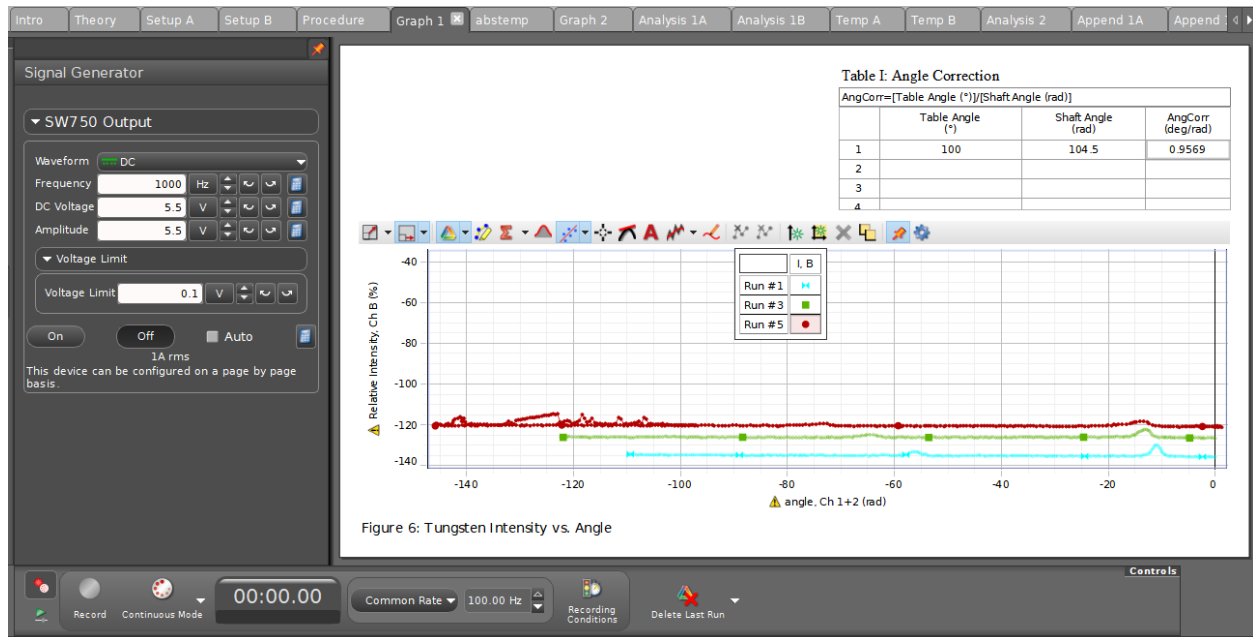


3. On the side bar select Signal Voltage and set the DC Voltage to 5 V to start.



4. Set the table at the start position.

5. Press Record and rotate the table till it can't rotate anymore. Then press the stop button.



6. Increment the DC voltage by .5 and run again. Do this for a total of 5 runs.